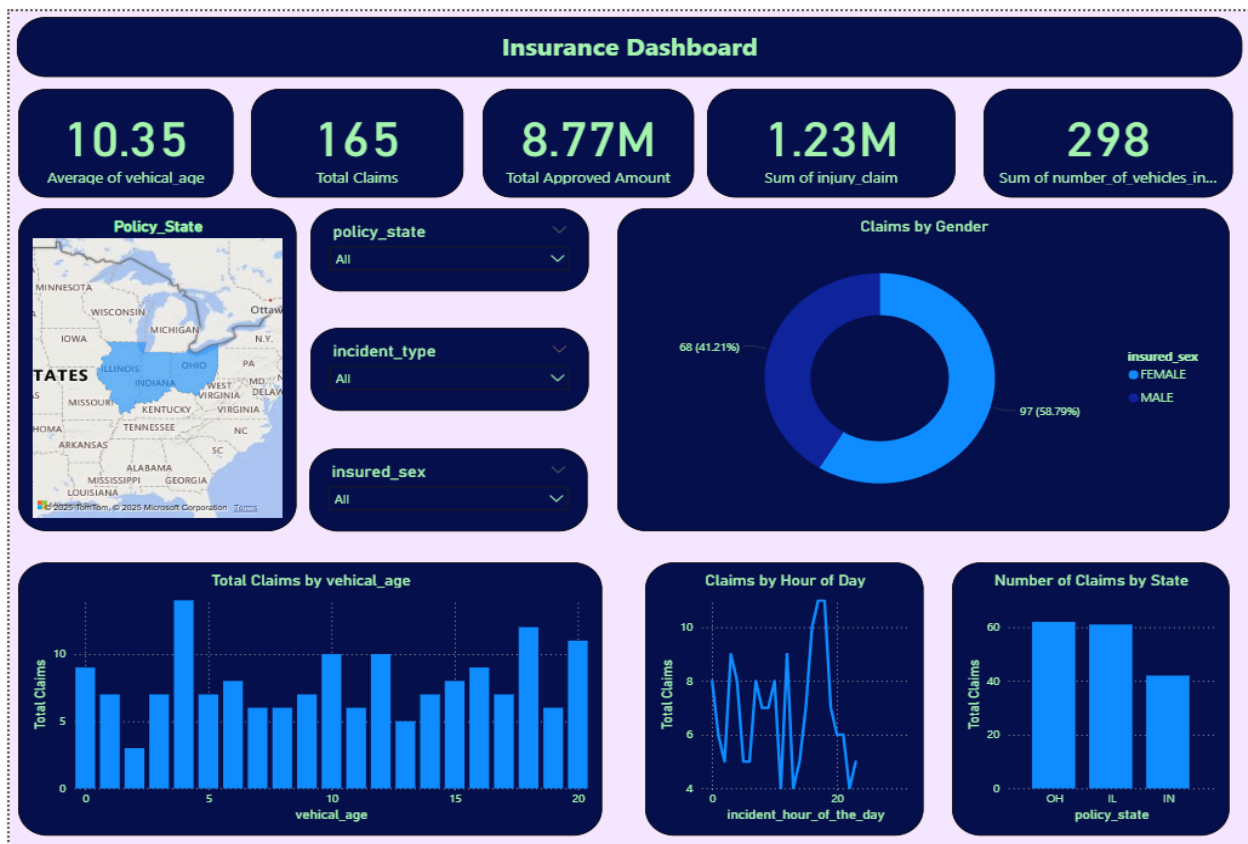


Insurance Dashboard

Power BI :

An Insurance company wants to visualize the financial performance of claims made against the policies. The data given contains the information about claims processed in the month of January and February. The aim is to plot visualisations which will provide a quick overview of claims data to the leadership.

Dataset Information: The data consists of 1000 individual claims. The most important variable of interest is fraud_reported. This variable is labelled 1 if a certain claim is reported to be fraudulent and 0 otherwise. Each claim in the data is described by 40 different attributes, which are represented as columns. We can divide the descriptive attributes into four main categories: the insured person, the policy of that person, description of the incident and characteristics of the car which is involved in the incident.



Future Analysis:

The objective of this project is to perform a **future-oriented analysis** of an insurance company's claim data to help management understand trends, identify fraud risks, and forecast future financial performance. The dataset provided consists of **1,000 individual claims** processed in the months of January and February. Each record represents a unique insurance claim and contains approximately **40 descriptive attributes**. These attributes include detailed information about the **insured person**, their **policy details**, the **incident description**, and the **vehicle characteristics** involved in the claim. The key variable of interest is **fraud_reported**, which indicates whether a claim was fraudulent (1) or not (0).

Using **Microsoft Power BI**, an interactive and visually engaging **dashboard** was created to analyze the company's operational performance and provide predictive insights for future planning. The project begins with importing and cleaning the dataset in Power BI. Data fields were categorized into meaningful groups such as Customer, Policy, Vehicle, and Incident information. Relationships and data types were verified to ensure accurate aggregation and analysis. Several **DAX measures** were then created to calculate total claims, profit margins, fraud ratios, and policy performance. One of the main DAX formulas used was:

$$\text{Profit} = \text{SUM}(\text{insurance}[\text{policy_annual_premium}]) - \text{UM}(\text{insurance}[\text{total_claim_amount}])$$

Q1. Various factors like number of claims processed and the total amount approved across various states.

The analysis shows that states like **California**, **Oregon**, and **Nevada** have the highest number of claims processed and the largest total claim amounts. These states contribute the most to the company's payout due to higher traffic density and severe accident rates. States with fewer claims show lower approved amounts, indicating better claim control. This highlights that claim volume and approved amount are strongly influenced by the **state's population, driving conditions, and risk exposure**.

Q2. Count of incidents and impact of time, gender and vehicle age.

The analysis indicates that **incident counts peak during late evening hours (around 5 PM–9 PM)** when traffic is heaviest. **Male policyholders** report a slightly higher number of incidents compared to females. Additionally, **older vehicles (especially those manufactured before 2010)** are linked to higher claim amounts and more severe accidents. This suggests that **time of day, driver gender, and vehicle age** significantly influence the likelihood and severity of insurance incidents.